# MUSHFIQUE TANZIM MUZTABA

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**MECHANICAL DESIGN ENGINEER | SOLIDWORKS | FEA | DESIGN FOR MANUFACTURE**

## PROFESSIONAL SUMMARY

Mechanical Engineering graduate from The Hong Kong Polytechnic University with practical experience designing mechanical components and products in SOLIDWORKS. Project work spans crash safety equipment at TGGS Bangkok, consumer product redesigns, and a final year autonomous robotics platform, covering concept development, CAD modelling, engineering drawings, and design iteration. Seeking mechanical design, R&D, or engineering roles in Hong Kong.

## EDUCATION

**The Hong Kong Polytechnic University** — **Hong Kong**
*Bachelor of Engineering (Honours) in Mechanical Engineering* — *Graduation Date: May 2026*

- **Accolades:** The Hong Kong Polytechnic University Entry Scholarship (Academic)
- **Relevant Coursework:** Fluid Mechanics, Engineering Thermodynamics, Numerical Methods for Engineers, Linear Systems and Control, Manufacturing Fundamentals, Fundamentals of Robotics

## TECHNICAL SKILLS

- **CAD and Mechanical Design:** SOLIDWORKS (modelling, FEA, assemblies), AutoCAD
- **Programming:** Python, C++, MATLAB
- **Robotics:** ROS2 (Humble, Jazzy), Gazebo, SLAM Toolbox, Cartographer, Nav2
- **AI and Computer Vision:** PyTorch, YOLOv11, OpenCV
- **Embedded Systems:** Raspberry Pi, STM32, I2C, SPI, GPIO
- **Engineering Tools:** Microsoft Power Apps, Power Automate, Git, Office 365

## WORK EXPERIENCE

**Alpha Vision Technology Limited** — **Hong Kong**
*Mechanical Design Intern* — *June 2026 – August 2026*

- Designed and additively manufactured the complete outer enclosure for a pipeline inspection crawler robot in SOLIDWORKS, taking the design from concept through functional prototype.
- Designed and 3D printed a controller enclosure with an integrated display opening, PCB mounting standoffs, and cable routing features to support reliable assembly and maintenance.
- Designed a multi-joint mechanical robot arm for a consumer humanoid robot in SOLIDWORKS, optimizing the structure for manufacturability, assembly, and aesthetic integration.

**MangDang Technology Co., Limited** — **Hong Kong**
*Programmer (Part-time)* — *February 2026 – March 2026*

- Migrated the CHAMP quadruped robot software stack from ROS 2 Humble to Jazzy, including simulator transition from Gazebo Classic to Gazebo Harmonic and full Ubuntu 24.04 compatibility, submitted to the public client repository as the agreed deliverable.
- Re-engineered the CHAMP quadruped contact sensor node from Gazebo transport APIs to GZ Harmonic, improving gait control accuracy.
- Validated autonomous navigation by testing SLAM (Cartographer, SLAM Toolbox), Nav2, and teleoperation on both simulator and Raspberry Pi hardware.

**CLP Power Hong Kong** — **Hong Kong**
*Sandwich Intern* — *July 2024 – June 2025*

Project Execution Team (PET)

- Built a Power Apps 'Lessons Learned' portal for 100+ members across 4 business groups to centralize and share project insights, with planned wider rollout.
- Supported project governance activities and completed CLP Project Management Academy training.

Hong Kong Battery Energy Storage System (HKBESS)

- Developed a Power Apps Management-of-Change tool for the Hung Shui Kiu “A” substation, centralizing engineering change requests.
- Documented 15+ operational risks for the battery storage system, informing project safety protocols.

**Thai German Graduate School of Engineering (TGGS)** **Bangkok, Thailand**
*Research Intern* *August 2023*

- Designed a motorbike crash-test dolly in SOLIDWORKS from concept sketches to production-ready assemblies.
- Performed iterative FEA under simulated crash loads, optimizing geometry for target safety factors and material efficiency.
- Engineered an energy-absorbing crash-cushion system, reducing peak impact forces and improving test repeatability.

## PROJECT EXPERIENCE

**Autonomous Indoor Thermal Inspection Robot** **January 2026 – May 2026**

- Developed an autonomous indoor inspection robot using ROS 2, SLAM Toolbox, and frontier-based exploration with LiDAR, RGB, and thermal sensors for non-contact anomaly detection (>35°C).
- Built a multi-layer perception pipeline combining thermal hotspot detection, object detection (YOLO11s), and Gemini 2.5 Flash multimodal analysis on a Raspberry Pi 5 with distributed processing via rosbridge.
- Implemented a natural-language control interface (Gemini 2.5 Flash) and operator dashboard (React) with live SLAM map, camera feeds, and thermal visualization.

**Autonomous Vision-Guided Targeting System** **November 2025**

- Developed an autonomous targeting system in Python/OpenCV on Raspberry Pi, achieving 90% detection accuracy under variable lighting via HSV colour thresholds.
- Implemented PID controller (Kp=0.12, Ki=0.005, Kd=0.25) with frame smoothing and dead zones for precise 2-axis servo control via I2C PWM.
- Integrated trajectory compensation and GPIO-controlled firing logic for reliable target acquisition at 1 m.

**PolyU E Formula Racing Team** **September 2023 – May 2024**

- Optimised suspension geometry using Lotus Suspension Analysis to evaluate handling characteristics for the team's electric formula vehicle.
- Designed tie rods, steering column, and steering linkages in SOLIDWORKS, ensuring manufacturability.

**Automated Material-Handling Trolley** **September 2023**

- Designed and manufactured an automated industrial trolley in SOLIDWORKS with motorized drive and steering mechanisms.
- Implemented autonomous loading/unloading, straight-line tracking, and 90° turning within a fixed-footprint environment.

**Table Fan Component Redesign** **November 2022**

- Analyzed and redesigned table fan components for improved performance and manufacturability, grounded in market and competitive research.
- Produced hand sketches, full SOLIDWORKS CAD models, and stress simulation across the assembly.

## LICENSES AND CERTIFICATIONS

- Hong Kong Green Card (Construction Industry Safety Training) *Valid until March 2028*
- Hong Kong Blue Card (Shipboard Cargo Handling Basic Safety Training) *Valid until January 2028*

**LinkedIn Learning Certificates:** SOLIDWORKS: Advanced Engineering Drawings, SOLIDWORKS: Advanced Simulation, SOLIDWORKS: Modelling Gears, Learning AutoCAD 2024.

## LEADERSHIP AND ACTIVITIES

**PolyU ENGL English Debate Club** **Hong Kong**
*Creative Designer* *August 2024 – June 2025*

- Designed posters, and social media graphics, strengthening brand engagement for 500+ members.

**PolyU International Student Association** **Hong Kong**
*Creative Designer and Social Media Manager* *July 2023 – June 2024*

- Created promotional graphics and videos garnering 15,000+ views, boosting event participation.

**PolyU Student Halls of Residence** **Hong Kong**
*Event Organizer* *November 2022 – June 2025*

- Organized and coordinated inter-hall activities for 120+ participants, fostering community engagement.